Planning

The Wilds Summer 2025 Field Work

Goal: Collect a multi-modal animal ecology dataset with overlapping spatial-temporal resolution at The Wilds.

The long-term goal is to use the dataset to build a digital twin of The Wilds to assist in planning multi-modal sensor deployments and test conservation strategies. We will publish the dataset to support the development of multi-modal AI models for biodiversity and environmental monitoring. We invite collaborators to identify biological research questions that can be answered with a multimodal dataset

Intro: The Wilds is a 10,000-acre safari park located in southeastern Ohio. Managed by the Columbus Zoo, this former strip mine is now home to 30+ rare and endangered species from Africa and Asia, as well as an activeresearch facility focused on habitat restoration and native biodiversity conservation. This summer, ICICLE researchers are collecting a multimodal dataset focused on a 100-acre area used by Pere David's deer. The deer are equipped with GPS ear tags providing hourly location updates and will also be monitored using camera traps, bioacoustic sensors, and drones.

Project Description:
The initial data collection phase for multimodal sensor networks in field animal ecology studies focuses on testing the ICICLE cyber-infrastructure, which includes autonomous drone navigation and camera trap models. This study aims to explore the system characteristics required to deploy Edge AI for heterogeneous sensor networks specifically designed for real-time, unsupervised adaptive sampling in the field. Our goal for this summer fieldwork project is to understand the strengths and weaknesses of each sensor modality in tracking land use, animal interactions, and individual identification. This will inform future multimodal sensor deployments for autonomous, adaptive ecosystem monitoring. Additionally, we hope this dataset will be valuable to the AI community as a training resource for developing models capable of synthesizing multiple data modalities and to the biological community for methods development.

This project marks the first step in establishing The Wilds as a biological research station and living laboratory for research at the intersection of technology and nature. **We invite your input on ecological research questions that can be explored using our multi-modal dataset, as well as suggestions for sensor deployment strategies that will maximize the project's value for research, both the ecological and AI research communities.**

Deployment Plan

Spatial: concentrate on a single pasture for proof-of-concept

https://earth.google.com/earth/d/10GVFBTCYF9KAoQrGtzBVqn38AuPclY2s?usp=sharing

Temporal: overlap sensor deployments so data is collected concurrently - ideally 5 concurrent days of data collection once in early summer (May/June) and once in late summer (August/Summer)

Sensor	Sampling Duration	Sampling Intervals	Manpower Required for Data Collection

Ear tags	Continuous for duration of study	Hourly pings with individual id and GPS locations	Assistance from the Wilds to access data feed
Quadcopter Drones	2-3 days, 2 hours each day (ideally AM and PM sampling)	2-3 times per week	Drone pilot + field assistant + The Wilds animal welfare expert
Fixed Wing Drones	1 session, 2 hrs approx	Once	Drone pilot + field assistant + The Wilds animal welfare expert
Satellite	Continuous for duration of study	Continuous	None
Camera trap	1 week	*Need confirm if photo or video and if motion-activated	Installation, battery replacement, SD card collection
Bioacoustic Monitors	1 week	Continuous	Installation, battery replacement, SD card collection
Weather station	Continuous	Continuous	Access to data from The Wilds

Sensors

Sensor	Data Description	Species	Research Questions	Current Data Availability	Hardware Availability
Ear tags	Hourly pings with individual id and GPS locations	Deer, bison	Land-use of wildlife Ground-truth for other modalities	Rolling deployment to ungulate herds	The Wilds rolling out to managed species
Quadcopter Drones	Videos	Zebras, giraffes, onagers, horses, AWD; others possible	Behavior studies, movement tracking	Existing research approval for select ungulates; autonomous and manned missions	ICICLE and Imageomics owned drones - Parrot Anafi, DJI mini, Sentinel
Fixed Wing Drones	Photos of wildlife and landscape; LiDAR data for 3D	Bison, others are possible	Land use of animals, presence of plant species, and detection of invasive species	Existing dataset of fixed-wing photos of bison; more possible	Dr. McCrink's lab has gliders
Satellite	Photos of the Wilds over time	Plants	How has the landscape evolved throughout the restoration efforts?	Free photographs; higher-resolution available for purchase	Free, plus more available for purchase
Camera trap	Photos and videos of animals	Exotic species and natives	Land use, biodiversity, and estimates of animal presence	African Wild Dog datasets	Imageomics owns GoPros and trail cameras

Bioacoustic Monitors	Sound recordings	Native pollinators, birds, and insects; managed ungulate species (PD deer, takins, phorses)	Can we identify species based on vocalizations?	N/A	Kitzes Lab may have some spares https://www.kitzeslab.org/; or may need to purchase
Weather	Detailed weather	All	Context for land-use and behavior; given rain, temperature, etc	Existing dedicated weather station	Existing Vantage Pro2 Sensor Suite

Ear tags

Figure: Moovement Screenshots from Dan B.

Quadcopter Drones

https://imageomics.github.io/wildwing/

Fixed-wing Drones

Camera Traps

Bioacoustic Monitors

https://www.openacousticdevices.info/audiomoth

Tentative Fieldwork Dates

Google Calendar

Indoor Tests (Dreese Labs)

- Familiarize ourselves with the tools and data collection
- Practice installing sensors and retrieving data
- Dates: May 15, 2025 May 29, 2025

Fieldwork Week 1 (The Wilds): Jun 23, 2025 - Jun 26, 2025

Fieldwork Week 2 (The Wilds): Aug 4, 2025 - Aug 7, 2025

Equipment to Purchase

Category	Model	Price per unit	Count	Link	Ordered? If so - ETA
Drone	HAWK'S WORK F450 Drone Kit to build, DIY Quadcopter Set, Frame + Pixhawk + GPS + Power Module + ESC + Brushless Motor + Prop	$400	5	Amazon	

	+ Battery + RC Transmitter&Receiver + Accessory + Guild Book&Video(Latest Upgrade)				
Bioacoustic monitors	Songmeter Micros	$150	10	https://www.wildlifeacoustics.com/products/song-meter-micro-2	
Camera trap	Browning Defender Vision Pro Live (Cellular)	$200	1 to start (plus 10 if its good)	https://www.trailcampro.com/products/browning-defender-vision-pro-live-cellular?_pos=1&_sid=246b20c66&_ss=r	
Camera trap	Spartan GoLive 2 M (Cellular)	$250	1 to start (plus 10 if it works out)	https://www.trailcampro.com/products/spartan-golive-2-m?utm_source=lantern&utm_campaign=trail-camera-selection-guide	
Camera trap	Gardepro T5NG (Non-Cellular) with solar panel	$140	1 to start - non-cellular, basic model; solar recommended for long deployments	https://www.trailcampro.com/products/gardepro-t5ng-solar-panel-package	
Camera trap mount	Slate River Stealth Game Camera Mount	$20	3 for now, plus more if we need	https://www.trailcampro.com/products/slate-ri	

				ver-stealth-game-camera-mount	
Solar panel for cam trap	Browning Solar Panel Power Pack	$60	1 for now	https://www.trailcampro.com/products/browning-solar-panel-power-pack?_pos=1&_fid=da284f570&_ss=c	
Micro SD Cards for drones and camera traps	Sandisk (atleast 128 GB, prefer higher)	$20-$40	20	Amazon	

Note: want to check to make sure cellcular works in the chosen pasture, it should, but want to confirm

Camera trap guys

- Solar good for solar to save battery
- https://www.trailcampro.com/products/spartan-golive-2-m?utm_source=lantern&utm_campaign=trail-camera-selection-guide (more programmability)
- Thermal motion
- https://www.trailcampro.com/products/browning-defender-vision-pro-live-cellular?_pos=1&_sid=246b20c66&_ss=r
- Solar for sure, backup batteries
  https://www.trailcampro.com/products/slate-river-stealth-game-camera-mount
- 32 or 64 sandisk
- Lock boxes
- Cable lock
- Universal swivel mount
- Discounts on orders of **5 units of the same thing** is bulk (but no refunds on 5 or more, so recommend order a few online before bulk purchase)
- 3 days (free shipping)
- Day & nighttime
- Ideally streaming back to cloud, or local network
- Wifi and cellular
-

landon@trailcampro.com

To Do List

- [x] ~~Submit research permit to the wilds: 📷 Multimodal Research Request~~
- [x] ~~Confirm masters and undergrads available to assist with field work this summer - ask Chris about this~~
- [ ] Confirm fieldwork dates with the wilds
- [x] ~~Purchase equipment~~

Feedback from All-hands

- Deploy the camera traps in a way that would answer ‘where do animals spend their time’?
- Check out papers on land-cover class placement
- Proximity to water - gradient distance to water
- Think about the ecology of the animal and what matters to them
- Account for variance and remove confounding variables
- Camera trap deployment for land use different than population estimates
- Stick with landuse, not population estimates for the managed species
- Ecological qs: how to their behaviors change throughout the season
    - Green wave - ndvi greenup
- Forage preference depending on wet to dry to season
- Do animals spend more time in the shade depending on the weather?
- How do the males’ behaviors change during rutting season? Where are they spending their time?

- 200 acre pasture
    - Used by PD deer (GPS tagged), also takin, phorses

- Bioacoustics
    - Ungulates make vocalizations - can we distinguish species using bioacoustic monitors?
    - PD deer will be in rutting season and make noises
    - Install AudioMoths near camera traps - have both visual and audio data

Resources

Reference Materials

Recommended Readings

Overviews
Towards the fully automated monitoring of ecological communities
Perspectives in machine learning for wildlife conservation

Datasets
MammAlps: A multi-view video behavior monitoring dataset of wild mammals in the Swiss Alps
The PanAf-FGBG Dataset: Understanding the Impact of Backgrounds in Wildlife Behaviour Recognition
https://github.com/agentmorris/agentmorrispublic/blob/main/drone-datasets.md
https://agentmorris.github.io/camera-trap-ml-survey/
https://www.movebank.org/cms/movebank-main
https://lila.science/

Tools
https://acoupi.github.io/acoupi/
https://birdnet.cornell.edu/
https://github.com/google-research/perch
MegaDetector

My papers:
Characterizing and Modeling AI-Driven Animal Ecology Studies at the Edge
WildWing: An open-source, autonomous and affordable UAS for animal behaviour video monitoring
MMLA: Multi-Environment, Multi-Species, Low-Altitude Drone Dataset

Animal behavior from drones
Advancing animal behaviour research using drone technology
Opportunities and risks in the use of drones for studying animal behaviour
Quantifying the movement, behaviour and environmental context of group-living animals using drones and computer vision

Existing Datasets

Our Datasets

https://huggingface.co/datasets/imageomics/KABR
https://imageomics.github.io/mmla/

Collections of other's datasets

https://bkellenb.github.io/cv-for-wildlife-aerial-imagery/
https://www.movebank.org/cms/movebank-main
https://lila.science/

Tools

https://github.com/Imageomics/wildwing/tree/main
https://github.com/KevynAngueira/SoftwarePilot
https://github.com/agentmorris/MegaDetector
https://github.com/google-research/perch

Start here for intro to computer vision: https://docs.ultralytics.com/

Data Management

Instructions

All data collected in the field (including Tuttle Park tests) will be saved to the Staging Server. Once initial analysis and organization is completed in the staging server, data will be transferred to OSC for permanent storage. We will use OSC resources to train models.

Example from previous NEON/Imageomics/ABC project:
NEON Protocol and Data naming conventions

Naming convention

location_MMDDYY/sensor_type/data

Staging Server

Server IP: 149.165.150.157
User: smartfield
Password: @1drivenAE
Storage location: /media/volume/generativedata/smartfields

To access from the command line terminal:

1. '*ssh smartfield@149.165.150.157*'
2. Enter password when prompted '*@1drivenAE*'

Transfer data

Rsync

Rsync tutorial: https://samuelstevens.me/writing/rsync

From command line: *rsync -avz /local/path/ user@remote-server:/remote/path/*

Example command: *rsync -avz /Volumes/NO NAME/DCIM smartfield@149.165.150.157:/media/volume/generativedata/smartfields/tuttlepark_052925/cameratrap*

Globus

See CVAT Globus instructions: CVAT Annotation Documentation

Permanent Data Storage: OSC

https://www.osc.edu/
Imageomics Project: PAS2136
Location: /fs/ess/PAS2136/TheWilds2025

Bioacoustic Project (Tanishka)

Bioacoustic Project (Tanishka)
The Wilds

BirdNET

BirdNET is an AI-powered system developed by the Cornell Lab of Ornithology that uses deep learning to analyze audio recordings and identify bird species based on their vocalizations. It supports a wide range of bird species and is widely used in ecological research, conservation efforts, and citizen science.

Use Cases at The Wilds:

- Studying Migration Patterns: Track how birds move over time by analyzing seasonal audio recordings.
- Assessing Habitat Quality: Identify bird species present in a habitat to evaluate biodiversity.
- Monitoring Populations: Track presence/absence trends of specific bird species over time.

PERCH

Perch is an open-source AI tool by Google for identifying animal sounds. It supports 10,000+ species and lets researchers train custom models for audio analysis on edge devices.

Use Cases:

- **Custom Species Detection**: Train models to identify rare or local bird calls not in BirdNET.
- **Call Density Estimation**: Measure how often species vocalize in different areas or time periods.
- **Multi-Species Detection**: Analyze birds, frogs, and insects using a single pipeline.
- **Smart On-Device Filtering**: Use Perch on Jetson Nano or Pi to log only relevant sounds.

BirdNET and PERCH Integration at The Wilds

1. **Audio Collection**
    a. Deploy bioacoustic sensors across different habitats.
    b. Record continuously or schedule (e.g., 5 minutes per hour).
2. **Initial Species Detection (BirdNET)**
    a. Use BirdNET for quick, on-device identification of known bird species.
    b. Ideal for real-time feedback and field monitoring.
3. **Advanced Analysis (Perch)**
    a. Apply Perch to rare, ambiguous, or behavior-specific audio.
    b. Generate embeddings, estimate call densities, or create custom classifiers.
4. **Data Integration and Visualization**
    a. Merge BirdNET results (species, timestamps) and Perch outputs
    b. (embeddings, call types).

Paper 1: Characterizing and Modeling AI-Driven Animal Ecology Studies at the Edge

- Runtime Adaptation: Using on-device AI to filter noise, adjust camera angles, or redirect drones improves data quality and speeds up insight generation.

- Resource Planning: Helps researchers choose the right edge setup before field deployment.

- Scalability: The framework can simulate larger deployments virtually—important for ethical and logistical reasons.

- Standardization: Introduces common SLO metrics to benchmark ADAE system performance.

Data Analysis

Data Analysis and Visualizations

Project scope

- Build integration between different platforms for each modality (APIs) with central repo
- Work with Matt Thompson and Elizabeth on this
- Modality - Platform
    - GPS data & drone data - Movebank https://www.movebank.org/cms/movebank-main
    - Camera trap - Wildlife Insights https://www.wildlifeinsights.org/
    - Bioacoustics - (TBD - Jenna will confirm)
    - Citizen science - iNaturalist https://www.inaturalist.org/
- Data movement
    - OSC <→ Movebank/iNat/WildlifeInsights <→ GoogleEarth

Visualization Project Requirements

1. Date, time, location, sensor modality, and observations in tabular format
    a. Observation of a particular individual at a particular time and location (who, where, when)
    b. Metrics for observations = count of animals by species
    c. Support to add more observation details later (example: sex, age, behavior)
    d. Convert annotated camera trap, acoustic, drone, and citizen science data to tabular format (team to provide annotated data)
    e. Join GPS data so it can be cross-referenced with other modalities
2. **Visualization of observations temporally and spatially**
    a. Google Earth might be a good place to start: The Wilds Google Earth
    b. See MoveBank as an example: https://www.movebank.org/cms/webapp?gwt_fragment=page=search_map (ideally we would want a small, MVP version of this)
    c. Want visualization to link back to sub-directory in HuggingFace (organized by modality and time range) SmartWilds HF
    d. Goal: website or dashboard to make it easy for ecologists and computer scientists who are interested in using the data to quickly see all the data available, and drill down by modality/time/observation
3. Analysis of strengths and weaknesses of each modality in capturing specific events (like deer crossing lake) and general land-use patterns for the deer.
    a. Goal: quantify the relative strengths and weaknesses of the different sensing modalities
    b. See metrics below: spatial, temporal, detection/species, efficiency
    c. Radar charts might be a good option to visualize these

Metric	Description	Examples / Notes
Spatial Range	Maximum area over which the sensor can collect data without relocation	GPS collars: entire home range; Camera traps: fixed location; Drones: up to battery/flyable distance
Spatial Granularity / Resolution	How finely the sensor can localize animal presence or behavior	GPS: ~1–10 m; Camera traps: fixed field-of-view, can detect exact position in frame; Drones: sub-meter resolution in imagery
Temporal Range / Deployment Duration	How long the sensor can function unattended	GPS: months–years; Camera traps: weeks–months; Drones: hours per flight; Bioacoustics: weeks–months depending on battery
Temporal Granularity / Sampling Rate	Frequency of data collection	GPS: hourly; Camera traps: event-triggered or time-lapse; Drones: video at 30–60 fps; Bioacoustics: continuous or triggered recordings
Species Detectability	Which animals can be reliably detected	Cryptic species (bioacoustics), large ungulates (camera traps/drones), wide-ranging (GPS)
Behavior Detail	Level of behavioral information obtainable	Drones: posture, interactions; Camera traps: limited to frame; Bioacoustics: vocalizations; GPS: movement patterns only
Ease of Use / Operator Effort	Human time required for deployment, monitoring, and maintenance	Drones: high; Camera traps: low–medium; GPS: low once deployed; Bioacoustics: low–medium
Energy Efficiency	Battery life or energy consumption per unit time	Drones: limited by flight time; GPS collars: months; Camera traps: weeks–months; Bioacoustics: weeks
Storage Capacity / Data Volume	Amount of data generated and storage needed	Drones/video: high; Camera traps: moderate; GPS: low; Bioacoustics: moderate–high

Summary Metrics:
Coverage Efficiency = (Spatial Range × Temporal Range) / Operational Effort
Data Resolution Index = Combined measure of spatial + temporal granularity
Species Coverage Score = Number of species detectable / effort required

Tools used for data annotation by modality

Modality	**Team Member**	**Tools**
Camera Traps	Bharath	SAMv2, Megadetector
Drone videos	Anirudh, Vedant, Rugved	OWL
Acoustics	Tanishka	Perch, BirdNet
GPS	Ally	TBD
Citizen science	Jenna	Megadetector, BioClip